Ex

$$x(t) = e^{j(t+\pi/4)} = e^{jt} \cdot e^{j\pi/4}$$

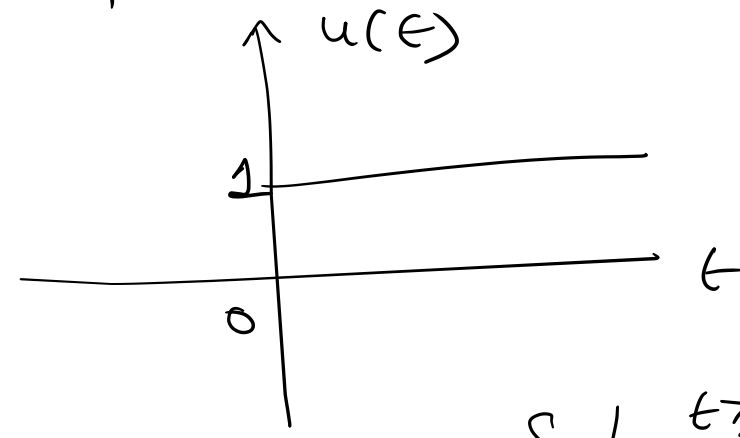
$\omega = 1$

$$e^{jt} = e^{j\omega t} = e^{j\omega(t+T)}$$

$$= e^{j\omega t} \, e^{j\omega T} = 1$$

$$\omega T = 2\pi K$$

$$T = 2\pi$$



$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

1.6 (a)

$$x_1(t) = x(t)\,u(t)$$

$$= e^{j(t+\pi/4)} \cdot u(t)$$

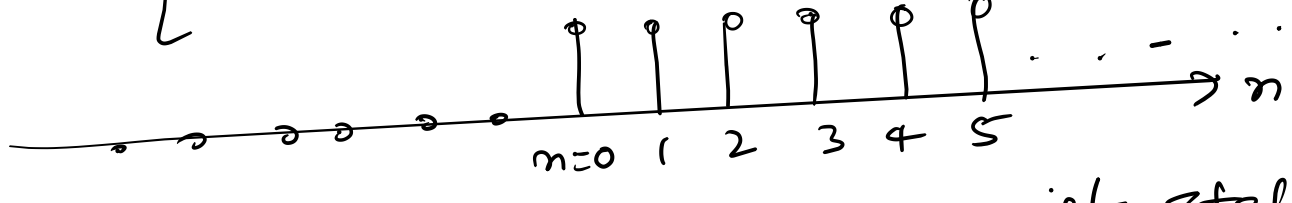
$$x(t) = \begin{cases} 0 & ; \quad t < 0 \\ e^{j(t+\pi/4} & ; \quad t \geq 0 \end{cases}$$

non-periodic

(b) $x_2[n] = u[n] + u[-n]$

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

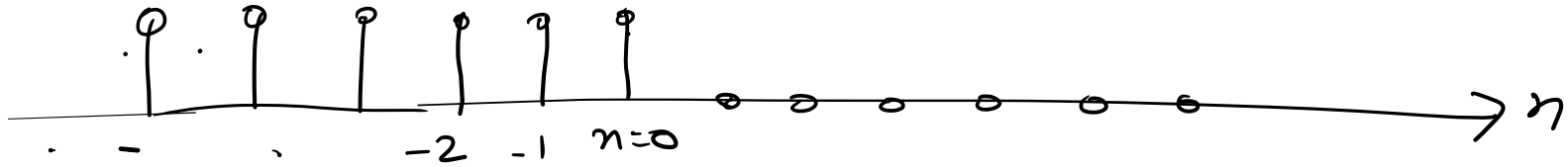
Unit Step sequence



D.T. unit step fn.

$u[-n] =$

$u[-n]$



variable

C.T. t ( )

$x(t)$ $t \in \mathbb{R}$

D.T. n [ ]

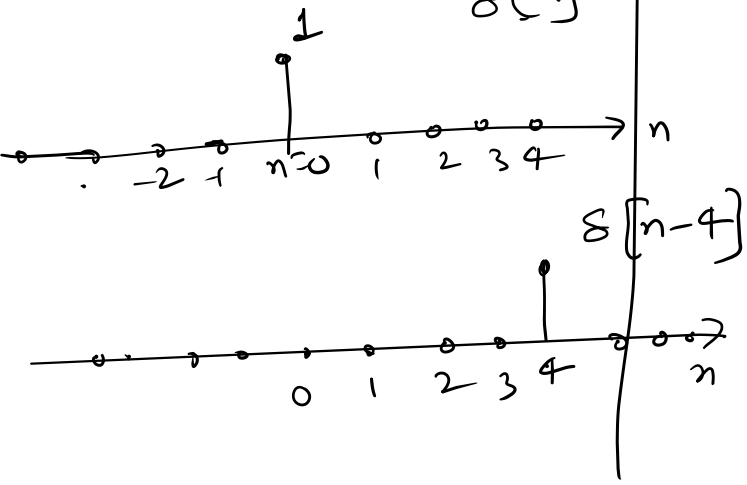
$n \in \mathbb{Z}$

$x[n]$

(c) $x_3[n] = \sum_{k=-\infty}^{\infty} \left(\delta[n-4k] - \delta[n-1-4k] \right)$

$\delta[n]$ = unit impulse fn.

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$



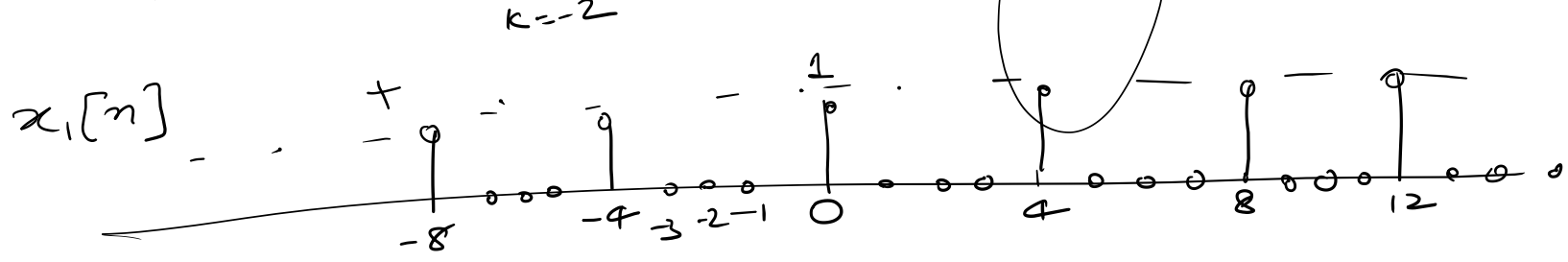
$$= \sum_{k=-\infty}^{\infty} \delta[n-4k] - \sum_{k=-\infty}^{\infty} \delta[n-1-4k]$$

$$= x_1[n] - x_2[n]$$

$$x_1[n] = \sum_{k=-\infty}^{\infty} \delta[n-4k]$$

$$= \ldots + \delta[n+8] + \delta[n+4] + \delta[n] + \delta[n-4] + \ldots$$

($k=-2$, $k=-1$, $k=0$, $k=1$)



1.8

Express the real part of the signals in the form of

$$A e^{-at} \cos(\omega t + \phi) \qquad A \geq 0 \;\&\; -\pi < \phi \leq \pi$$

(a) $x_1(t) = -2$

$$= 2 e^{j\pi} = 2\cos\pi + j\sin\pi$$

$$= 2 e^{0.t} \cos(0.t + \pi)$$

$A = 2$, $a = 0$, $\omega = 0$, $\phi = \pi$

(b) $x_2(t) = \left(\sqrt{2}\right)\left(e^{j\pi/4}\right)\left(\cos(3t + 2\pi)\right)$

(c) $x_3(t) = e^{-t}\sin(3t + \pi)$

$$= e^{-t}\sin\left(3t + \pi/2 + \pi/2\right) = e^{-t}\cos\left(3t + \pi/2\right)$$

$A = 1$, $a = +1$, $\omega = 3$, $\phi = \pi/2$

(d) $x_4(t) = j\, e^{(-2+j100)t}$

$= e^{j\pi/2} \cdot e^{-2t} \cdot e^{j100t}$

$= e^{-2t} \, e^{j(100t + \pi/2)}$

$\mathrm{Re}\{x_4(t)\} = 1 \cdot e^{-2t} \cos(100t + \pi/2)$

$A = 1, \quad a = 2, \quad \omega = 100, \quad \phi = \pi/2$

1.9 (a)

Find periodic or not. if periodic, Specify the fundamental time period.

$$x_1(t) = j e^{j10t}$$

$$= e^{j(10t + \pi/2)}$$

if is periodic $x(t+T) = x(t)$ $\forall t$

if smallest +ve T that satisfies this eq. is the fundamental time period.

$$x(t+T) = e^{j(10t + \pi/2 + 10T)} = \underbrace{e^{j(10t+\pi/2)}}_{x(t)} \cdot e^{j10T}$$

if $e^{j10T} = 1$, then the smallest T for which this holds is the fun. time period.

$$10T = 2\pi$$

$$T = \frac{2\pi}{10} = \text{F.T.P.}$$

(b) $x_2(t) = e^{-(-1+j)t} = e^{t} \cdot e^{-jt} = e^{t}(\cos t - j \sin t)$

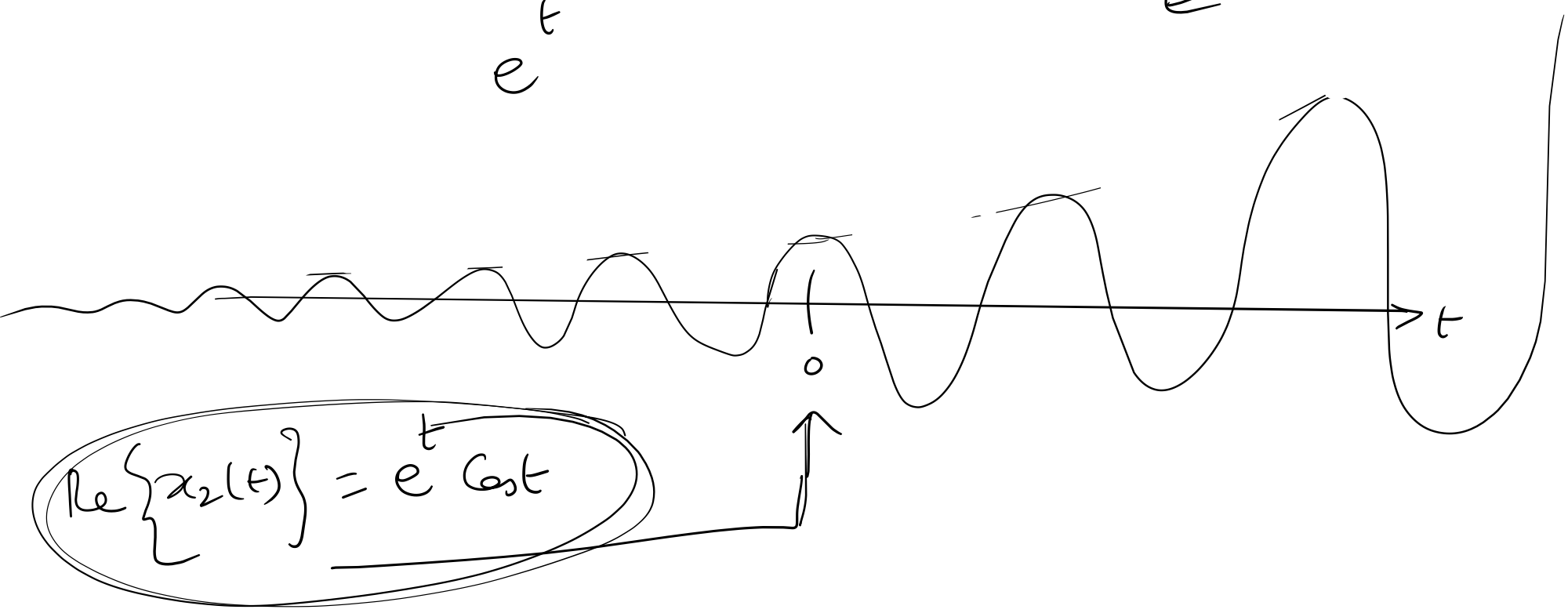
$= y_1(t) \cdot y_2(t)$

periodic with F.T.P

$T = \frac{2\pi}{|\omega|}$

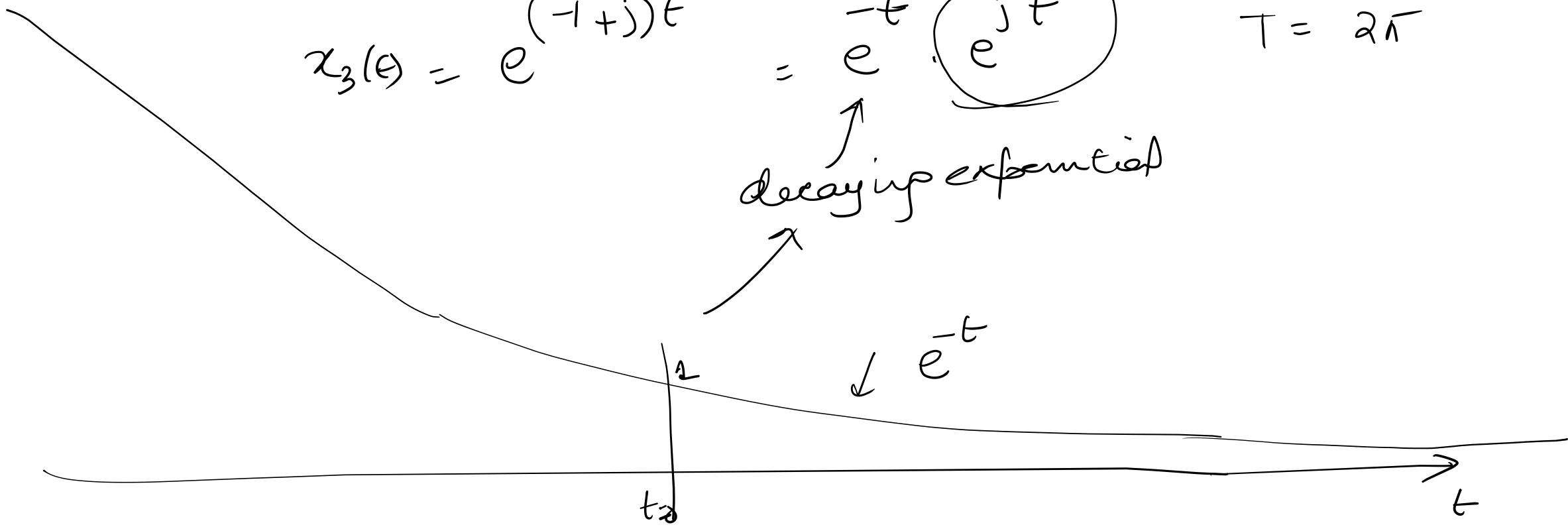
$= 2\pi$

Not periodic



$$x_3(t) = e^{(-1+j)t} = e^{-t} \cdot e^{jt} \qquad T = 2\pi$$

decaying exponential



Signal is periodic if $x_3(t+T) = x_3(t) \quad \forall t$

$$x_3(t+T) = e^{-(t+T)} \cdot e^{j(t+T)}$$

$$= \left(e^{-T} \cdot e^{jT}\right) \cdot e^{-t} \cdot e^{jT} \cdot x(t)$$

$\Longleftarrow$ 1

22/8/2025

Class

Quiz Syllabus

Examples of Chapter 1 &

Q. 1 to Q. 34

D.T. Signals

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

D.T. Unit Step fn.

or Unit Step Sequence

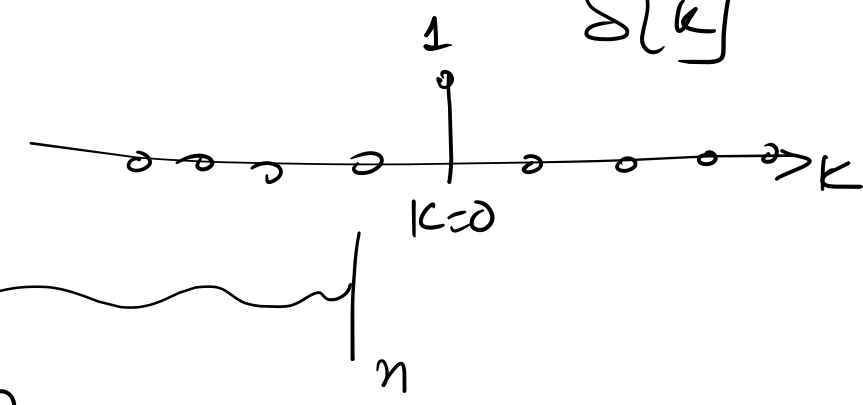
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

D.T. Impulse fn.

$$x[n] = u[n] - u[n-1] \equiv \text{First difference}$$

$$= \delta[n]$$

$$\delta[n] = u[n] - u[n-1]$$



$$\boxed{u[n] = \sum_{k=-\infty}^{n} \delta[k]}$$

$$u[-1] = \sum_{k=-\infty}^{-1} \delta[k] = 0$$

$$u[-2] = \sum_{k=-\infty}^{-2} \delta[k] = 0$$

$$u[0] = \sum_{k=-\infty}^{0} \delta[k] = 0 \cdots + \underset{\uparrow}{1} = 1$$
$$k=0$$

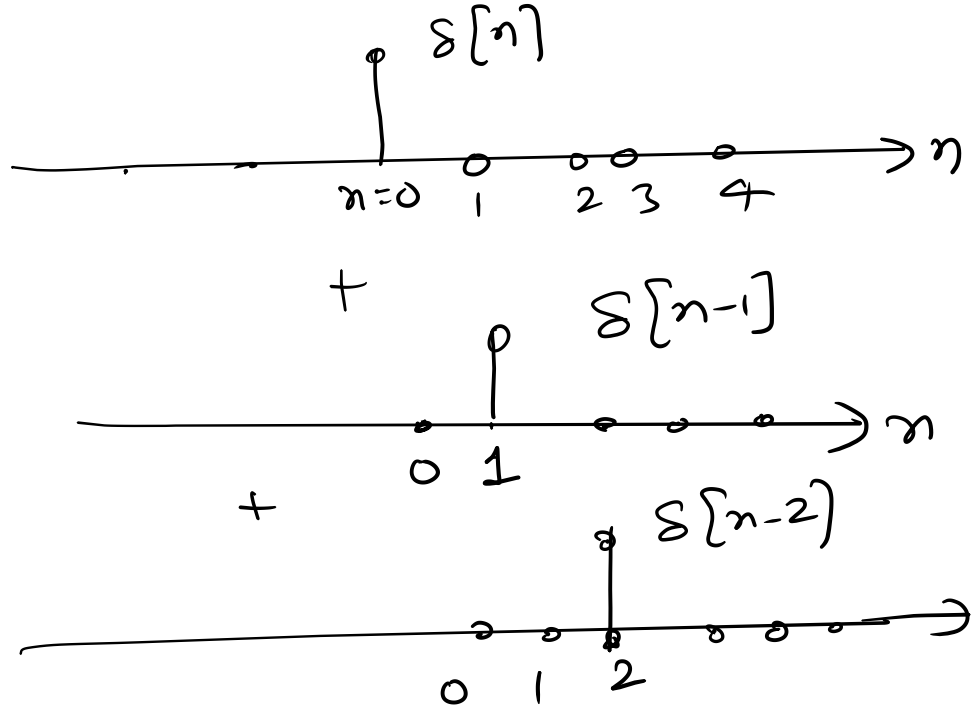
$$u[1] = \sum_{k=-\infty}^{1} \delta[k] = 0 \cdots 0 + \underset{\uparrow}{1} + \underset{\uparrow}{0} = 1$$
$$k=0 \qquad k=1$$

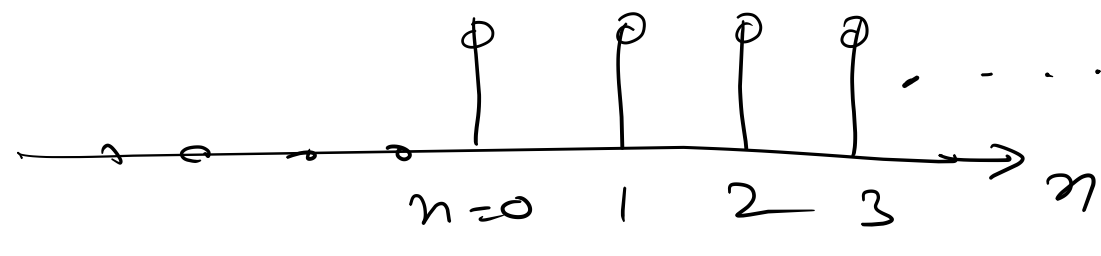
$$u[n] = \sum_{k=-\infty}^{n} \delta[k] = 1$$

$n > 0$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

$$= \delta[n] + \delta[n-1] + \delta[n-2] + \cdots$$



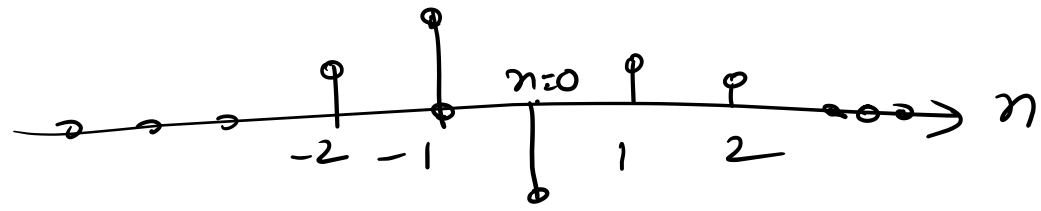


$u[n]$

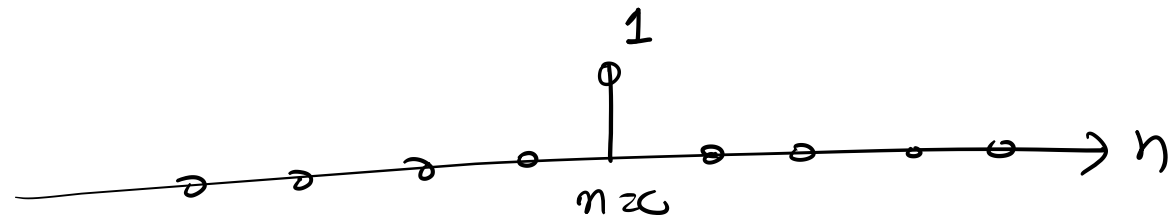
$$y[n] = x[n]\,\delta[n]$$

Is this statement $y[n] = x[0]\,\delta[n]$ correct?

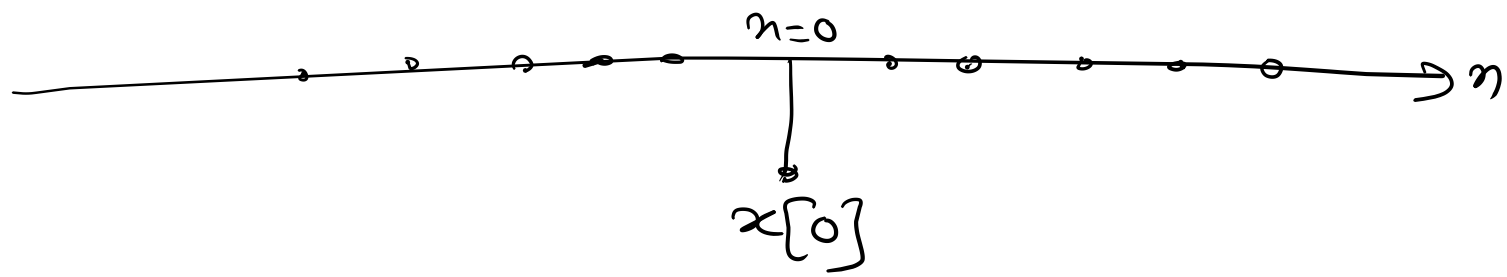
Say, my $x[n]$



$\delta[n]$



$x[n]\,\delta[n]$



$= x[0]\,\delta[n]$

$$y[n] = x[n]\,\delta[n-n_0]$$

$$\delta[n-n_0] = 1$$

when $n - n_0 = 0$

$n = n_0$

$$= x[n_0]\,\delta[n-n_0]$$

$\because n_0 =$ constant

$x[n_0]$ → constant

$\delta[n-n_0]$ → fn. of the variable n.

I have sampled $x[n]$ at $n = n_0$

Also called as Sifting of signal $x[n]$ at $n = n_0$

~~$\delta(t) = 1 \quad t=0$~~ C.T. signals

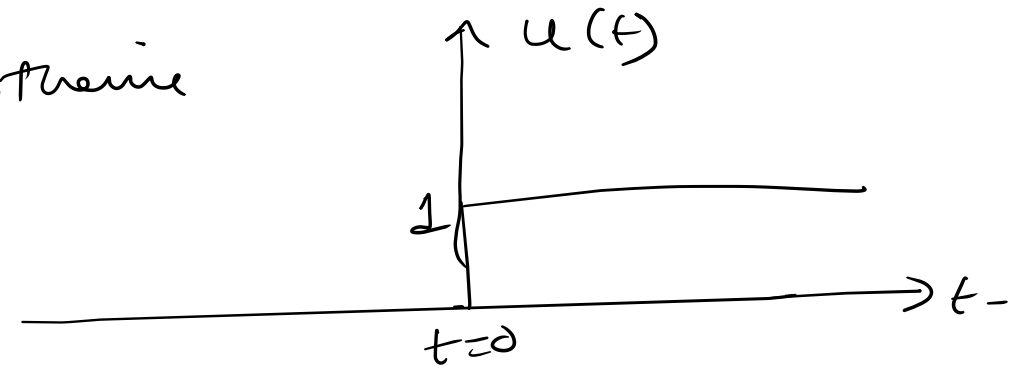
$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

& $\delta(t) = 0 \quad t \neq 0$

Unit step fn in C.T.

$t \geq 0$

$$u(t) = \begin{cases} 1 \\ 0 \end{cases}$$ otherwise



$$u(t) = \lim_{\Delta \to 0} u_\Delta(t)$$

$$\delta_\Delta(t) \triangleq \frac{d}{dt} u_\Delta(t)$$

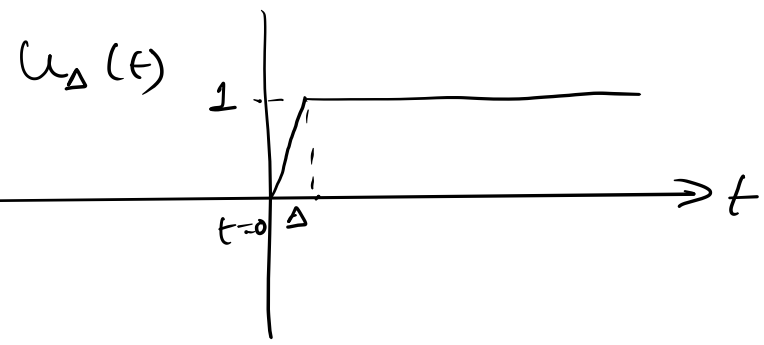


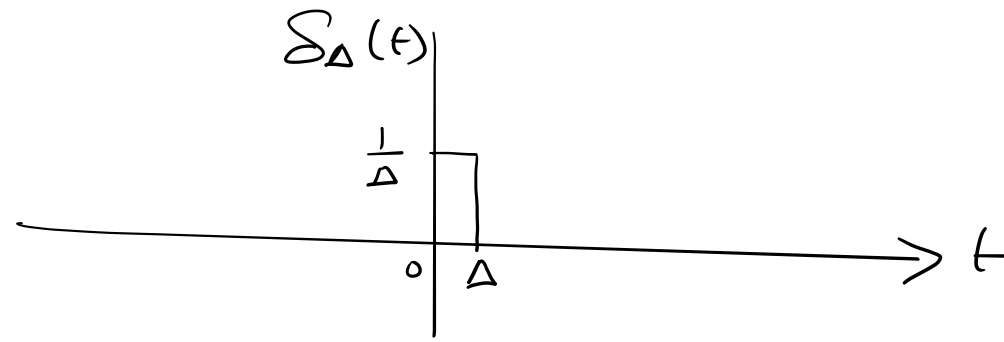
$$\int_{-\infty}^{\infty} \delta_\Delta(t)\,dt = 1$$

Define $\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t)$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

Unit Impulse fn.





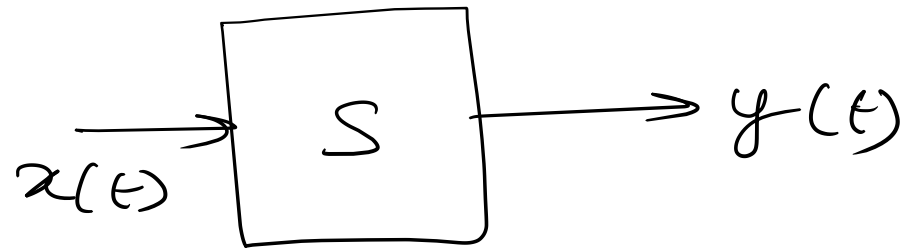
Check!

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

$$u(t) = \int_{0}^{\infty} \delta(t-\tau)\, d\tau$$

System properties

$$S: \; y(t) = S\{x(t)\}$$



System with memory

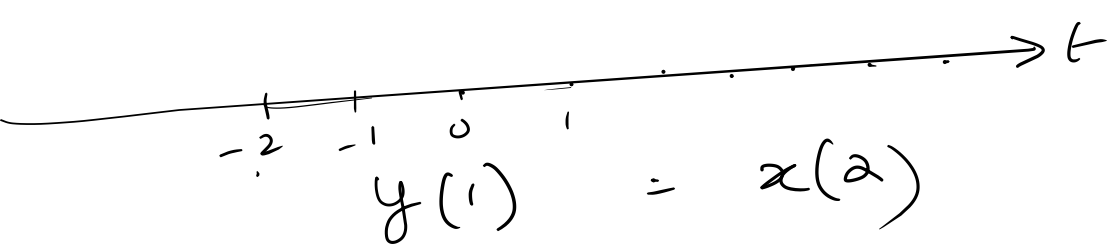
C.T.

S: $y(t) = x(t-2)$ — w. memory

$y(t) = x(2t)$

$y(t) = \sin t \; x(t)$ — w/o memory

$y(-1) = x(-2)$



$y(1) = x(2)$

D.T.

S: $y[n] = x[n-2]$ — Systems with memory

$y[n] = x[2n]$

$y[n] = \sin(n)\, x[n]$ — w/o memory

$y(t) = x(t+1)$

System with memory

Causal Systems

Causality — Unless there is a cause, there will not be an effect o/p

For a causal system, o/p depends on present & past i/ps.

Examples of causal systems

$y(t) = x(t)$

$y(t) = x(t-1)$

$y(t) = \sin t\, x(t-1)$

$y(t) = x^2(t)$

$y(t) = t\, x(t)$

$y[n] = x[n]$

$y[n] = x[n-1]$

$y[n] = \sin n\, x[n-1]$

$y[n] = x^2[n]$

$y[n] = n\, x[n]$

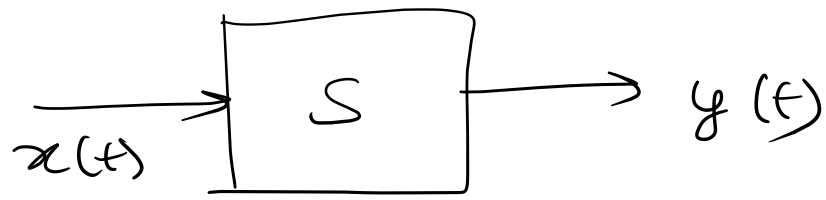
non-causal

$y(t) = x(t+2)$

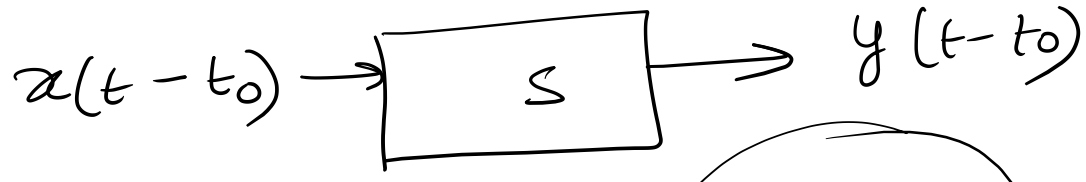
$y[n] = x[n+2]$

$y[n] = \frac{1}{3}\{x[n+1] + x[n] + x[n-1]\}$

Time - varying or Time - invariant Systems



Time - invariant



$x_1(t) \longrightarrow y_1(t)$

Time - variant: $x_1(t - t_0) \longrightarrow y_1(t - t_0)$

If $x_2(t) = x_1(t - t_0)$, $x_2(t) \longrightarrow y_2(t)$

$y_2(t) = y_1(t - t_0)$

If this property is not satisfied, Then the System is time - varying

$$y(t) = x(2t)$$

$x_1(t) \longrightarrow y_1(t) = x_1(2t)$

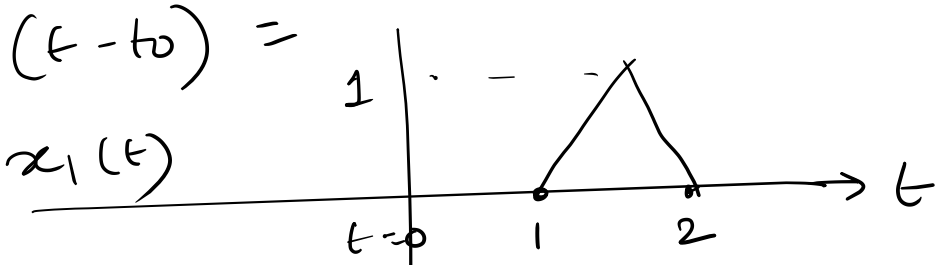
if $x_2(t) = x_1(t - t_0)$

& $y_2(t) = y_1(t - t_0)$

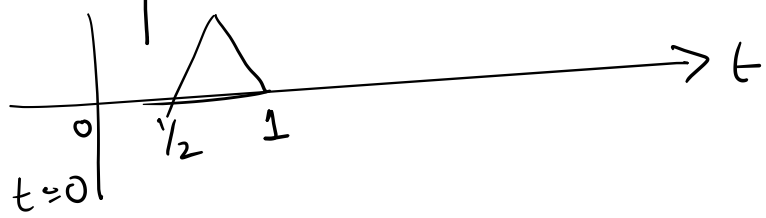
$y_1(t) = x_1(2t)$

$t_0 = 2$

$y_1(t - t_0) =$



$y_1(t) =$



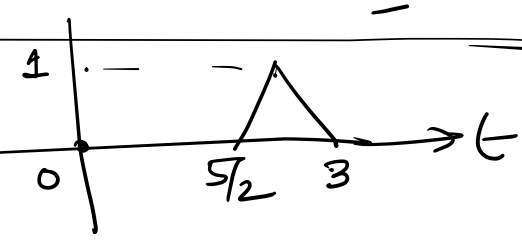
$x_2(t) \longrightarrow y_2(t) = x_2(2t)$

, then the system is Time Invariant

$x_2(t)$
$x_1(t - t_0)$ $\longrightarrow$ $y_2(t) = x_2(2t)$

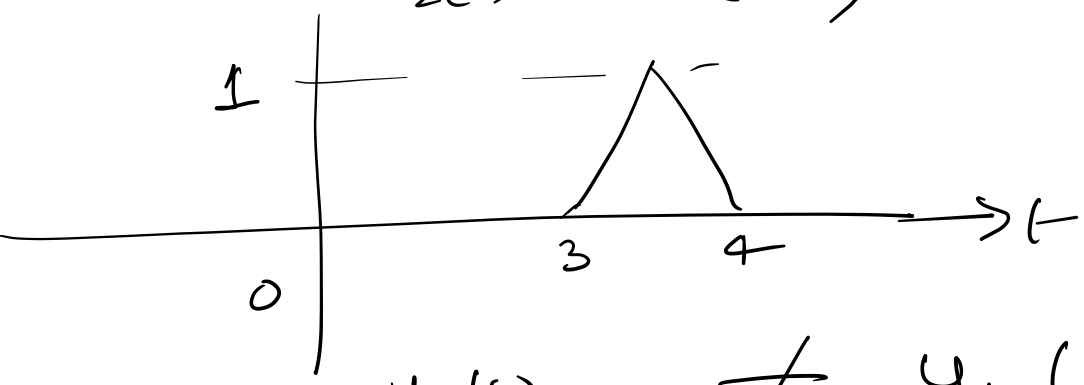
$=$

$y_1(t - 2) =$

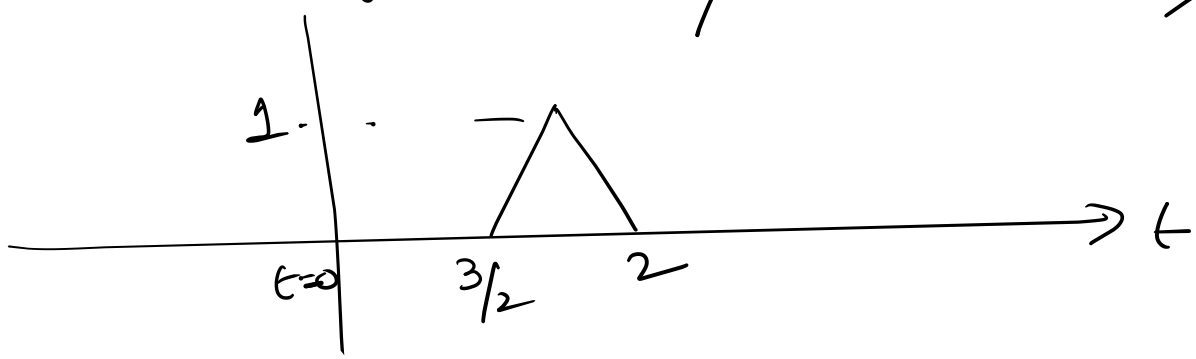


First Delay

$x_2(t) = x_1(t-2)$



$y_2(t) \neq y_1(t-t_0)$



$x_2(t) = x_1(t-2)$

$\downarrow$

$y_2(t) = x_2(2t)$

$x_1(t-2) \longrightarrow x_1(2t-2)$

$\uparrow$

$y_1(t) = x_1(2t)$

$y_1(t-t_0) = x_1(2t-2t_0)$